UNIVERSITI TEKNOLOGI MALAYSIA

SECP3623 - DATABASE PROGRAMMING

2025/2026 - SEM 1

Project II

Name	**:**	**JOANNE CHING YIN XUAN**
Matric Id	:	A23CS0227

Name	**:**	**CHUA JIA LIN**
Matric Id	:	A23CS0069

Name	**:**	**EVELYN GOH YUAN QI**
Matric Id	:	A23CS0222

Name	**:**	**LIM YU HAN**
Matric Id	:	A23CS0241

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PRESENTATION SLIDE:

EVENT MANAGEMENT & TICKET BOOKING SYSTEM

Using MongoDB Backend

Innovating Solutions

Case Study Introduction

This project implements a NoSQL environment for the Event Management and Ticket Booking System using MongoDB. The system is used to manage users, events, ticket bookings, and users reviews. The system supports operations such as registering new users, creating new events, booking tickets, and submitting feedback.

In this project, the system uses collections, documents, arrays, and references to store data flexibly.

Objectives

1. Create a NoSQL data model to manage users, events, bookings, and reviews.
2. Implement CRUD operations (Create, Read, Update, Delete) for all collections.
3. Demonstrate advanced MongoDB operations (indexing, aggregation pipelines, and sorting)
4. Create meaningful queries for analytics.
5. Analyze NoSQL security considerations and design challenges.

NoSQL Modelling Explanation

- Document-based data model using MongoDB
- Main collections:
  - Users
  - Events
  - Bookings
  - Reviews
- Embedded documents:
  - User address
  - Event location
- Referenced documents:
  - userId and eventId in bookings and reviews
- Support flexible schema and scalability

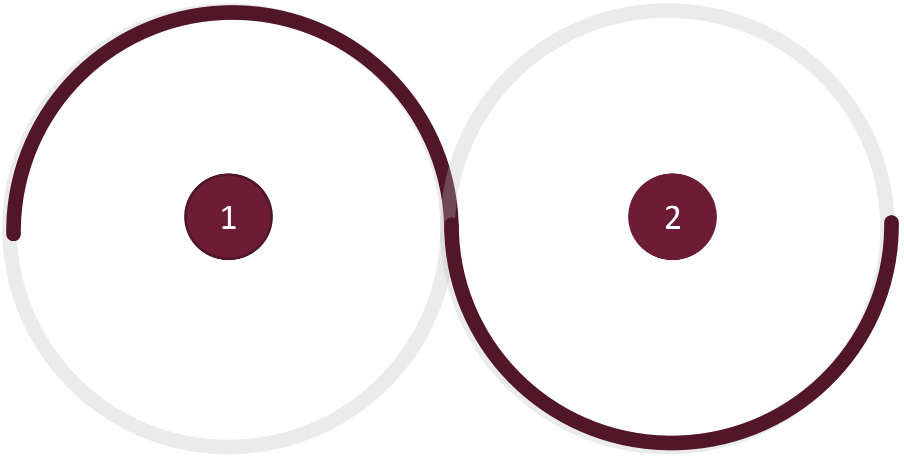
Basic MongoDB Operations (CRUD) demonstration:

1. Insert Operations
2. Query Operations
3. Update Operations
4. Delete Operations

This will be shown using MongoDB

Advanced MongoDB Operations :

Indexing



Aggregation Pipelines

This will be shown using MongdoDB

Team roles

Team Member	Role
Joanne Ching Yin Xuan	- Perform RUD operations in collections. - Perform aggregation and sorting operations in collections - Conclude what was learned, challenges, and recommendations for improvement - Justification of modifications
Chua Jia Lin	- Designed collections schema - Implemented CRUD and aggregation operations in collections - Analyzed the effect of indexing towards query performance - Introduced the topic chosen and explained the objectives of the project

Team roles

Team Member	Role
Evelyn Goh Yuan Qi	- Perform RU operations in collections - Perform indexing and aggregation operations in collections - Analyzed and documented the challenges encountered during database development. - Explained the security considerations and system limitations of the database design.
Lim Yu Han	- Perform RU operations in collections. - Implement bulk update operations to modify documents efficiently. - Demonstrated sorting operations to organize event records logically. - Contributed to the NoSQL data model explanation and documented query logic in the report.

THANK YOU

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